FORUM NOKIA

Push To Talk: Invitation Message Structure

Version 1.0; August 18, 2004

Push to Talk

NOKIA

Contents

Change History

August 18, 2004	Version 1.0	Initial document release

1 Introduction

Push to Talk over Cellular (PoC) introduces a direct one-to-one and one-to-many voice communication service in the cellular network. It makes a popular two-way radio service available through attractive mobile devices, thus enhancing cellular services and bringing new business opportunities into the domain of real-time voice communications.

Push to Talk service is a genuine differentiated voice service, because it is not a substitute for any existing cellular services. It gives operators an opportunity to develop their voice-service offerings without having to change conventional voice services. Third-party developers can benefit from this new type of voice service by creating different kinds of voice-based applications, such as dispatching and infotainment applications.

The PoC solution is based on half duplex Voice over IP (VoIP) technology over the 2G General Packet Radio Service (GPRS) and 2.5 Enhanced GPRS (EGPRS) networks. Building the service over the existing EGPRS network will enable fast service rollout, reduce the investments needed, and establish a natural growth path towards other radio access technologies, such as Wideband CDMA (WCDMA).

Thanks to IP technology, the Push to Talk service is “always on” and uses cellular access and radio resources more efficiently than circuit-switched cellular services. Network resources are thereby reserved only one-way for the duration of talk spurts instead of two-way for an entire call session. Among the many benefits of this solution over two-way radio systems are the coverage provided by the cellular network and the simple, fast creation of talk groups and group calls.

Note: This solution does not meet emergency public safety requirements!

1.1 Communication Principle

The principle of communication in PoC is simple — just push to talk. Users can select the person or talk group they wish to talk to, and then press the Push to Talk key (PTT) to start talking. The call is connected in real time. Push to Talk calls are one-way communication: while one person speaks, the other(s) only listen. Push to Talk speech is connected without the recipient(s) answering and heard through the device's built-in loudspeaker.

For one-to-many communication, users can create ad hoc talk groups, simply by sending invitations to the desired group members to join the talk group. Talk groups can also be preconfigured, for example, by the operator for the use of a specific group of people.

1.2 Technology

One of the basic functions of PoC is virtual connection within a group of people. For example, setting up a traditional conference call requires the user to make several phone calls; in PoC, the members of an active talk group session can converse in real time at the press of a key, even if conversation in the talk group session ended some time ago. This virtual connection state within a group session is created by using the Session Initiation Protocol (SIP).

PoC runs over the GPRS network and all the other GPRS services are available for PoC users.

2 Real-Time Group Call

A real-time group call means establishing real-time voice communication to a selected group by pressing a PTT key. Push to Talk calls are one-way communication: while one person speaks, the other members of the group session listen. The opportunity to speak is requested by pressing the PTT key and is granted on a first-come, first-served basis.

In order to be able to communicate in a group session, a user must join the session by selecting it from the device menu.

2.1 Provisioned Group

A provisioned group is a talk group that has been preconfigured to the system by the operator or another authorized management user. A closed user group and user rights to services define the group members. Provisioned groups are identified by URLs.

2.2 Ad Hoc Group

In the ad hoc group feature, the end users themselves can create the groups. Thus, a user does not have to have access to a preconfigured group in order to have group communication within the PoC service. From the network perspective, group definitions are temporary and will disappear in a predefined time from the call processor. However, a group may be stored in devices for further use. Ad hoc groups are identified by URLs.

3 Invitation Message Structure

An invitation message is sent over Short Message Service (SMS) from one device to another PoC device. This chapter describes all the mandatory and optional parameters the message can contain.

3.1 APPID: “w-nokia-poc-group” — Push to Talk Group Invitation

- **Parm: APPID:** Must be “w-nokia-poc-group”
- **Characteristic: RESOURCE:**
  - **Parm: URI:** Group Uri (PocGroupUri)
  - **Parm: NAME:** (PoCGroupNickname) Group name, optional
  - **Parm: FROM:** (PoCInviterUri) Inviter Uri
  - **Parm: FROMNAME:** (PocInviterNickname) Inviter's name, optional

3.2 XML Example

This XML example illustrates some parameters that the invitation message XML can contain.

```
<characteristic type="APPLICATION">
  <parm name="APPID" value="w-nokia-poc-group"/>
  <characteristic type="RESOURCE">
        <parm name="URI" value="Testi@poc.nokia.com"/>
        <parm name="NAME" value="Testi"/>
        <parm name="FROMNAME" value="Shura"/>
  </characteristic>
 </characteristic>
```

3.3 WBXML Conversion

XML to WAP Binary XML (WBXML) conversion is done according to the WBXML specification provided by the Open Mobile Alliance (OMA). Content must be converted “as it is” and no tokens should be used. This example shows how the XML document (see Section 3.2, “XML Example”) should be converted to Binary XML format.

```
Version (1.1)
01

Public Identifier (unknown)
01

Charset (UTF-8)
6A
```

```
String table length

00


Characteristic (with attributes and content)

CF 10 03 41 50 50 4C 49 43 41 54 49 4F 4E 00 01

   type, inline string, "APPLICATION\0", END


Parm (with attributes)

8E 0F 03 41 50 50 49 44 00 11 03 77 2D 6E 6F 6B 69 61 2D 70 6F 63 2D 67
72 6F 75 70 00 01

   name, inline string, "APPID\0", value, inline string, "w-nokia-poc-
group\0", END


Characteristic (with attributes and content)

CF 10 03 52 45 53 4F 55 52 43 45 00 01

   type, inline string, "RESOURCE\0", END


Parm (with attributes)

8E 0F 03 55 52 49 00 11 03 54 65 73 74 69 40 70 6F 63 2E 6E 6F 6B 69 61
2E 63 6F 6D 00 01

   name, inline string, "URI\0", value, inline string,
"Testi@poc.nokia.com\0", END



Parm (with attributes)

8E 0F 03 4E 41 4D 45 00 11 03 54 65 73 74 69 00 01

   name, inline string, "NAME\0", value, inline string, "Testi\0", END



Parm (with attributes)

8E 0F 03 46 52 4F 4D 4E 41 4D 45 00 11 03 53 68 75 72 61 00 01

   name, inline string, "FROMNAME\0", value, inline string, "Shura\0",
END



END characteristic

01



END characteristic

01
```

3.4 User Data Header (UDH)

In order to send the invitation to a PoC client, the User Data Header (UDH) needs to be in place. The UDH contains the concatenation information (if needed) and originating and destination ports. The receiving device always ignores the originating port address. In this example, the invitation SMS fits to one SMS message, so there is no need for concatenation. A single SMS message can hold up to 140 bytes of information. This message length is 140 bytes. The receiving end listens to port 49997, which is `c3 4d` in hexadecimal.

```
06 05 04 c3 4d c3 4d

Header specifying UDH length (number of octets)
06

Header specifying 16 bit port addresses
05

Header specifying IE length in bytes
04

Port, destination
c3 4d

Port, source
C3 4d
```

3.5 Complete Data for SMS

The UDH and WBXML example contains only the payload data that will be used by the PoC client at the receiving end. In order to send the content using SMS message delivery information such as the Short Message Service Center (SMSC) address, the recipient address and message type information need to be added to SUBMIT-PDU.

The TPDU parameter part should look like this:

0041000C9153484514701500 00A0

UDH:

060504C34DC34D

UD:

01016A00CF10034150504C49434154494F4E00018E0F034150504944001103772D6E6F6B69612D706F
632D67726F75700001CF10035245534F5552434500018E0F035552490011035465737469 40706F632E
6E6F6B69612E636F6D00018E0F034E414D4500110354657374690 0018E0F0346524F4D4E414D450011
03536875726100010101

4 Terms and Abbreviations

Term or abbreviation	Meaning
PoC	Push to Talk over Cellular
SMS	Short Message Service
WBXML	WAP Binary XML
UDH	User Data Header

5 References

WBXML *Binary XML Content Format*, WAP Forum, WAP-192-WBXML, http://www.openmobilealliance.org/tech/affiliates/wap/wapindex.html

Appendix A Registration Document for PoC Invitation Message

```
IDENTIFYING INFORMATION

#######################

APPID: wpocgroup.

APPID type: OMNA.

Owner: PoC Group Invite version 1.0.

Contact: Nokia.

Registration version: 1.

Registration timestamp: 2003-03-27.

Application description: Push to Talk group invitation.

Application reference: Reference to Forum Nokia/PoC.


WELL-KNOWN PARAMETERS

#####################

Characteristic/name: APPLICATION/APPID.

Status: Required.

Occurs: 1/1.

Default value: None.

Used values: "wpocgroup".

Interpretation: N/A.



Characteristic/name: APPLICATION/RESOURCE/URI.

Status: Required.

Occurs: 1/1.

Default value: N/A.

Used values: N/A.

Interpretation: Group Uri (PocGroupUri).

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Characteristic/name: APPLICATION/RESOURCE/NAME.

Status: Required.
```

Occurs: 0/1.

Default value: N/A

Interpretation: (PoCGroupNickname) Group name.

APPLICATION-SPECIFIC PARAMETERS

###############################

Characteristic/name: APPLICATION/RESOURCE/FROM

Status: Required.

Occurs: 0/1.

Default value: N/A.

Used values: N/A.

Interpretation: (PoCInviterUri) Inviter Uri.

Characteristic/name: APPLICATION/RESOURCE/FROMNAME.

Status: Required.

Occurs: 0/1.

Default value: N/A.

Used values: N/A.

Interpretation: (PocInviterNickname) Inviter's name.

PARAMETER VALUES

################

None.

EXAMPLE

#######

```
<characteristic type="APPLICATION">
  <parm name="APPID" value="w-nokia-poc-group"/>
  <characteristic type="RESOURCE">
        <parm name="URI" value="mygroup@someoperator.com"/>
```

```
        <parm name="NAME" value="My Group"/>
        <parm name="FROM" value="firstname.lastname@someoperator.com"/>
        <parm name="FROMNAME" value="Username"/>
  </characteristic>
 </characteristic>
###END###
```